

$$\frac{\theta_b}{\theta_s} = P(s) = \frac{K}{(s-p_1)(s-p_2)}$$

$$K = 138,3035$$

$$p_1 = -13,2515 \text{ rad/s} \rightarrow -2,1030 \text{ Hz}$$

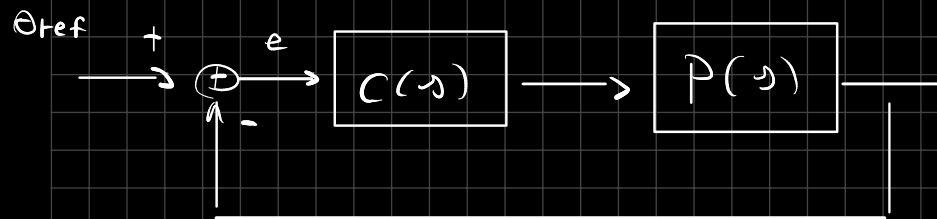
$$p_2 = -32,6059 \text{ rad/s} \rightarrow -5,1834 \text{ Hz}$$

$$\omega = 2\pi f$$

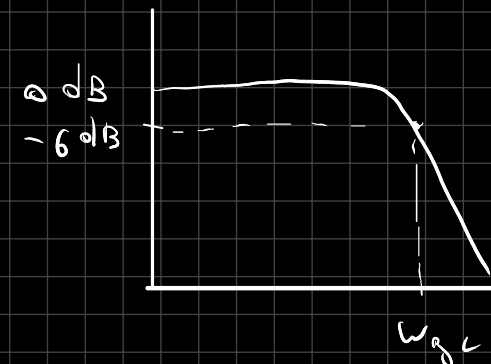
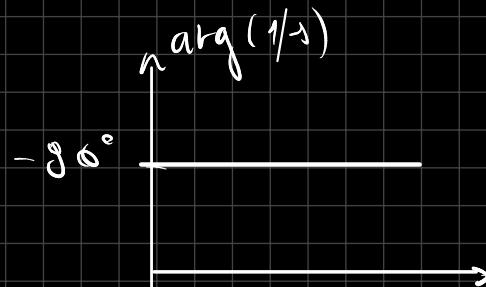
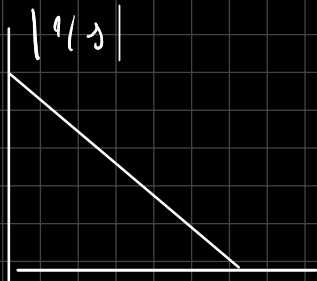
$$f_s = 50 \text{ Hz}$$

$$u = \theta_s$$

$$C(s) = \frac{q}{s}$$



$$C.P = \frac{q \cdot K}{s} \frac{1}{(s-p_1)(s-p_2)}$$



$$\omega_{gc} \sim BW_{CL}$$

$$T = \frac{L}{L+1} \Big|_{\omega=\omega_{gc}} = 1/2$$

$$\text{queremos } \omega_{gc} = 15 \text{ Hz} = 94,25 \text{ rad/s} \rightarrow \text{muy poco MF}$$

Para $MF = 20^\circ \rightarrow q = 20 \text{ dB} = 10$ $\omega_{gc} = 3,13 \text{ rad/s} = 0,5 \text{ Hz}$

$$\frac{e}{r} = S(s) = \frac{1}{1+L} \quad \frac{u}{r} = C \cdot S = \frac{C}{1+L} \rightarrow \text{Todo em ordem} \checkmark$$

$$C(s) = \frac{10}{s} \quad \text{com transform} \rightarrow \frac{2}{T_s} \frac{z-1}{z+1}$$

$$C(z) = 10 \cdot \frac{T_s}{2} \frac{z+1}{z-1} = 0,1 \frac{z+1}{z-1} = \frac{u(z)}{e(z)}$$

$$\rightarrow 0,1 \cdot (z+1) e(z) = (z-1) \cdot u(z) \quad \text{Prop: } x(n-k) \rightarrow z^{-k} \cdot X(z)$$

$$\rightarrow 0,1 \cdot z e(z) + 0,1 e(z) = z \cdot u(z) - u(z)$$

Divido por z

$$0,1 \cdot e(z) + 0,1 z^{-1} e(z) = u(z) - z^{-1} \cdot u(z)$$

$$z^{-1}$$

$$\rightarrow 0,1 e_k + 0,1 e_{k-1} = u_k - u_{k-1}$$

$$\rightarrow u_k = 0,1(e_k + e_{k-1}) + u_{k-1}$$

$$e = \theta_{ref} - \theta_b \quad u = \theta_s$$

$$\theta_s(k) = 0,1 \left(2\theta_{ref} - \theta_b(k) - \theta_b(k-1) \right) + \theta_s(k-1)$$

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